Demystifying Linux

A beginner's guide to installing, setting up and using Ubuntu 12.04

Ever overheard geeks talk about 'Linux' or 'Open Source Software' and wonder what they meant? Fear not, we're here to familiarize you with Linux and introduce you to a whole new way of interacting with computers. Before we get our hands dirty though, let's go over a few basics.

The Operating System (OS)

The OS is the underlying software that allows other programs such as your web browser, word processor and video player to run. Microsoft Windows is an OS, as is Apple's Mac OS X. At the heart of every OS lies a 'kernel' that tells a computer how to interact with various hardware devices. Linux is an Open Source kernel. Wondering what that means? Read on to find out!

Open Source

Software is written as a set of instructions telling the computer what to do. These instructions are referred to as 'source code'. If the source code is freely available for modification and redistribution, the software is called Open Source. Linux is developed and maintained by a large community of contributors, thanks to its open source nature.

Linux Distribution

Any OS that uses the Linux kernel is a Linux 'Distribution', or 'distro' in short. There are thousands of such distros available, each suited to a specific purpose.

Why should you use Linux?

GNU/Linux presents many significant advantages. Firstly, most distros are free of cost, and acquiring one is as easy as downloading an image and burning it to a disk (If you're a regular reader, you'll have noticed that we bundle at least one distro with every DVD.). The sheer number and variety of distributions out there will ensure that you'll find one to match your need. We'll be using Ubuntu as an example, known for its ease of use and beginner-friendly approach, not to mention a massive repository of applications and excellent community support.

Machines running Linux are unparalled on the performance and stability front. While Windows users may have to deal with system crashes (the infamous 'Blue Screen of Death') or frequent slowdowns, Linux's stability is legendary, and many of the world's biggest companies rely on servers running Linux to keep their data safe. There even exist lightweight Linux distros for very old and outdated computers, giving them a new lease of life!!

While Windows (and more recently, Mac OS X) users are frequently the subject of malware attacks, Linux is an inherently secure system. The number of Linux-specific viruses and trojans are very few, and a Windows virus is very often ineffective against a machine running Linux.

Lastly, Linux distros can do everything that other OSes can do. Watching movies or listening to music? Done. Browsing the web, checking your email or editing documents and spreadsheets? Done. Distributions like Ubuntu achieve all this and much more with ease, requiring minimal effort on the part of the user.

A word of caution though: Linux uses slightly different ways of doing things than what you're used to, so there may be a learning curve involved. If you run into problems or need help, remember that chances are someone else has had a similar problem before and you'll find the solution online.

Now that we've got that covered, let's get started!

As stated before, we'll run you through the steps required to obtain a copy of Ubuntu, create a bootable disk/pen drive, install Ubuntu on your system and customize it to your needs. We've included relevant screenshots wherever necessary to help you out, so the whole process should be a cakewalk.

1: Obtain the Ubuntu image

The DVD accompanying this issue (titled '*nix and Open Source') carries an Ubuntu 12.04 image. If you don't have the DVD, head over the http://www.ubuntu.com/download/desktop and download either the 32-bit or 64-bit version of the .iso file, depending on your system type. If you're unsure, go with the 32-bit one. You can also find torrent links on the site if you prefer using a BitTorrent client.

Step 2: Create a bootable disk/pen drive

The .iso file that you've downloaded will be used to create a bootable disk. A bootable disk is one that lets you boot directly from the device, rather the hard disk. If you decide to use a CD-ROM, use a tool like ImgBurn or PowerISO to burn the file to a disk. If you're using a pen drive (at least 1 GB) use a utility like UNetBootin (see image) to create a bootable pen drive.

3. Boot from the live CD/pen drive

Use UNetBootin to create a bootable pen drive

Restart your computer, keeping the pen drive plugged in or the CD in the CD-ROM drive. As soon as the computer starts to boot, press F12 or F2 (The exact key may vary depending on your BIOS. Look for a line at the bottom of the screen informing you of the same) to